

Multiple Choice (10 marks)

1. Which of the following most likely explains why Venus does not have a strong magnetic field?
 - (a) Its rotation is too slow.
 - (b) It has too thick an atmosphere.
 - (c) It is too large.
 - (d) It does not have a metallic core.
2. Life on Earth originated from
 - (a) We don't know for sure.
 - (b) aliens (panspermia).
 - (c) comets.
 - (d) the Earth's oceans.
3. How do astronomers think Jupiter generates its internal heat?
 - (a) nuclear fusion in the core
 - (b) by contracting changing gravitational potential energy into thermal energy
 - (c) internal friction due to its high rotation rate
 - (d) chemical processes
4. What type of radiation causes a black hole to evaporate over time?
 - (a) Schwarzschild radiation
 - (b) Planck radiation
 - (c) Kolmogorov radiation
 - (d) Hawking radiation
5. Which planet(s) in the solar system has/have active plate tectonics?
 - (a) Mars
 - (b) Venus
 - (c) Earth
 - (d) Mars and Earth

True / False (10 marks)

Answer True or False

6. True or False: The two moons of Mars are named Phobos and Deimos.
7. True or False: The atmosphere on Mars is too thin to trap a significant amount of heat, limiting the greenhouse effect's effectiveness.
8. True or False: The ability to breathe oxygen is not an essential requirement for life as we know it, as many organisms thrive without it.
9. True or False: The Andromeda Galaxy is located about 1.7 million light years away from Earth, making it one of our closest galactic neighbors.
10. True or False: Both Saturn and Jupiter have equatorial wind speeds that exceed 900 miles per hour, contributing to their dynamic atmospheres.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the current challenges faced by the Mars Exploration Rovers Opportunity and Spirit, focusing on their operational limitations and the implications for ongoing research.
12. Discuss the significance of the Arecibo Telescope in the field of radio astronomy, emphasizing its unique features and contributions to our understanding of the universe.

End of Paper